Lecture 23 2025-12-02

Forward/explicit Euler method

Approximations y_i (book $w_i \hat{=} y_i$)

hope $y_i \approx y(t_i)$ $\quad t_i = a + ih \quad i \in \mathbb{N}$

$$(FE) \begin{cases} y_0 = \alpha \\ y_{i+1} = y_i + h f(t_i, y_i) \quad i = 0, 1, 2, \dots \end{cases}$$

Test: $\begin{cases} y' = \lambda y \\ y(0) = \alpha \end{cases}$ exact solution $y(t) = \alpha e^{\lambda t}$

Error estimates

Preliminary lemmas

Lemma 5.7 For all $x \geq -1$ & m positive, $0 \leq (1+x)^m \leq e^{mx}$

Lemma 5.8 Let $s, t > 0$, $\{a_i\}_{i=0}^k$ sequence with $a_0 \geq -\frac{t}{s}$

& $a_{i+1} \leq (1+s) a_i + t$

Then $a_{i+1} \leq e^{(i+1)s}\left(a_0 + \frac{t}{s}\right) - \frac{t}{s}$

Proof: by induction (later)

Theorem 5.9 Suppose $f = f(t,y) \in C([a,b] \times \mathbb{R}^n)$ & Lipschitz continuous in y with Lipschitz constant L on $[a,b] \times \mathbb{R}^n$.

Suppose also that $\exists M$ with $|y''(t)| \leq M$ on $[a,b]$

(y solves (ODE): $y' = f(t,y)$, $y(a) = \alpha$)

Then the approximations $y_0, y_1, \dots, y_N$, $N \in \mathbb{N}$ generated by forward Euler method (FE) satisfy for $i = 0, 1, 2, \dots, N$

$$|y(t_i) - y_i| \leq \frac{hM}{2L}\left(e^{L(t_i - a)} - 1\right)$$

(goes to zero with $h \to 0$, $O(h)$)

Proof: $i=0$: clear since $y_0 = \alpha = y(a) = y(t_0)$

Taylor expansion: $y(t_{i+1}) = y(t_i) + h f(t_i, y(t_i)) + \frac{h^2}{2} \underbrace{y''(\xi_i)}_{|\cdot| \le M \text{ by assumption}}$ (1)

$i = 0, 1, \dots N-1$

(FE) $y_{i+1} = y_i + h f(t_i, y_i)$ (2)

Subtract (2) from (1)

$$y(t_{i+1}) - y_{i+1} = y(t_i) - y_i + h\left(f(t_i, y(t_i)) - f(t_i, y_i)\right) + \frac{h^2}{2} y''(\xi_i)$$

Take absolute values & use Δ-inequality for RHS

$$|y(t_{i+1}) - y_{i+1}| \le |y(t_i) - y_i| + h \underbrace{|f(t_i, y(t_i)) - f(t_i, y_i)|}_{\le L |y(t_i) - y_i| \text{ by Lipschitz continuity of } f} + \frac{h^2}{2} \underbrace{|y''(\xi_i)|}_{\le M}$$

$$\le (1 + \underbrace{hL}_{s}) \underbrace{|y(t_i) - y_i|}_{a_i} + \underbrace{\frac{h^2}{2} M}_{t} \quad \text{in Lemma 5.8}$$

$\Rightarrow$ By Lemma 58, we have

$$|y(t_{i+1}) - y_{i+1}| \le e^{(i+1)Lh} \left(\underbrace{|y(t_0) - y_0|}_{=0} + \frac{h^2 M}{2Lh} \right) - \frac{h^2 M}{2Lh}$$

$$\le e^{(i+1)Lh} \frac{Mh}{2L} - \frac{Mh}{2L} = \left(e^{(i+1)Lh} - 1\right) \frac{Mh}{2L}$$

$(i+1)h = t_{i+1} - a$

$$= \frac{Mh}{2L} \left(e^{L(t_{i+1} - a)} - 1\right) \quad \forall\, i = 0, 1, \dots N-1 \quad \blacksquare$$

Note:

- $y''(t) = \frac{dy'}{dt}(t) \overset{\text{(ODE)}}{=} \frac{d}{dt} f(t, y(t)) \overset{\text{chain rule}}{=} \partial_t f(t, y(t)) + \partial_y f(t, y(t)) \frac{d}{dt} y(t)$

$$\overset{\text{(ODE)}}{=} \partial_t f(t, y(t)) + \partial_y f(t, y(t)) f(t, y(t)) \quad \text{J}$$

$\Rightarrow$ If f has bounded 1st derivatives (& is bounded), assumption $|y''(t)| \le M$ is satisfied

- As $h \to 0$, the error goes to zero linearly for fixed t.
  (error $= O(h) \Rightarrow$ method is 1st order accurate)

- <u>Truncation error</u>: what remains if you plug in the exact solution into the scheme

$$\underbrace{\tau_{i+1}}_{\text{truncation error at } t_{i+1}} = \frac{y(t_i+h)-y(t_i)}{h} - f(t_i, y(t_i))$$

($y_i \neq y(t_i)$ in general, so τ_{i+1} is not $=0$!)

Also called "one step error"

When $|\tau_i| = O(h^p)$ we say scheme is (formally) p-th order accurate.

<u>Proof of Lemma 5.8</u>: By Induction

<u>Induction base</u>: <u>$i=0$</u>: $a_1 \leq (1+s)a_0 + t = \underbrace{(1+s)}_{\leq e^s \text{ by Lem 5.7}}\left(a_0 + \frac{t}{s}\right) - \frac{t}{s}$

$$\leq e^s\left(a_0 + \frac{t}{s}\right) - \frac{t}{s}$$

$\Rightarrow$ Claim true for $i=0$

<u>Induction step:</u> Assume true for i : $a_i \leq e^{is}\left(a_0 + \frac{t}{s}\right) - \frac{t}{s}$

$$a_{i+1} \overset{\text{assumption of Lemma}}{\leq} (1+s)a_i + t = (1+s)\left(a_i + \frac{t}{s}\right) - \frac{t}{s}$$

$$\overset{\text{Induction hypothesis}}{\leq} (1+s)\left(e^{is}\left(a_0 + \frac{t}{s}\right) - \frac{t}{s} + \frac{t}{s}\right) - \frac{t}{s}$$

$$= (1+s)e^{is}\left(a_0 + \frac{t}{s}\right) - \frac{t}{s}$$

$$\overset{\text{Lem. 5.7}}{\leq} e^s e^{is}\left(a_0 + \frac{t}{s}\right) - \frac{t}{s}$$

$$= e^{s(i+1)}\left(a_0 + \frac{t}{s}\right) - \frac{t}{s}$$

■

To generate higher order accurate methods, one could use Taylor's

theorem and add higher order terms from series.

Disadvantages: • higher order (partial) derivatives of f

$\Rightarrow$ complicated update formulas, especially for systems

• often numerically unstable for high order methods

see Section 5.3 of textbook

Therefore we often use Runge-Kutta Methods

5.4 Runge-Kutta Methods

Idea: Use several f evaluations to get from t_i to t_{i+1} but no higher order derivatives of f.

Examples: Forward Euler:

$$\begin{cases} k_1 = f(t_i, y_i) \\ y_{i+1} = y_i + h k_1 \end{cases}$$

Midpoint rule:

$$k_1 = f(t_i, y_i)$$

$$k_2 = f\left(t_i + \tfrac{1}{2}h, y_i + \tfrac{1}{2}h k_1\right)$$

$$y_{i+1} = y_i + h k_2$$

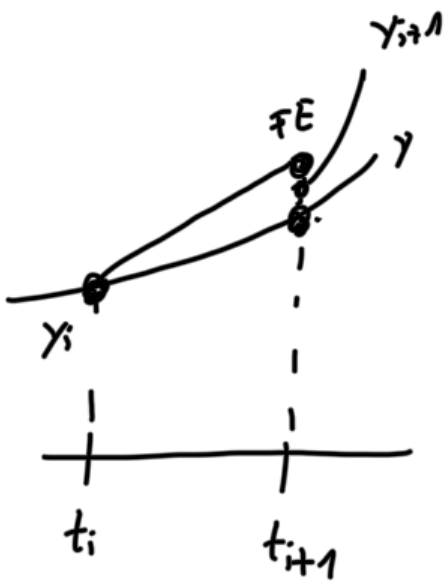
(book includes h in definition of the k's but this is not very common)

Modified Euler/explicit trapezoidal rule:

$$k_1 = f(t_i, y_i)$$

$$k_2 = f(t_i + h, y_i + h k_1)$$

$$y_{i+1} = y_i + \frac{h}{2}(k_1 + k_2)$$



These are 2-stage methods

General (explicit) s-stage method

$$k_\ell = f\left(t_i + c_\ell h, y_i + h \sum_{j=1}^{\ell} a_{\ell j} k_j\right)$$

$$y_{i+1} = y_i + h \sum_{\ell=1}^{s} b_\ell k_\ell$$

k_ℓ = ℓ'th stage derivative (book includes h with k)

$c_\ell, a_{\ell j}, b_\ell$ = scheme coefficients